Seeing the Same Task Differently

Vision-Language Model Operation, Benchmark Comparison, and a Qwen2.5-VL Fine-Tuning Case Study

A report built from the benchmark and fine-tuning artifacts stored in this repository

Primary analysis command:

```
python comparison\build_report.py -results-dir results -output-dir comparison\output -bootstrap-samples 5000
```

Chapter 1

Models

1.0.1 The Models Chosen

As mentioned in the report, the models chosen are:

Proprietary/API models

- `gpt-5.5`
- `gpt-5.4`
- `o3`
- `o4-mini`
- `claude-opus-4-8`
- `claude-opus-4-7`
- `claude-opus-4-6`
- `claude-sonnet-4-6`
- `claude-sonnet-4-5`
- `gemini-3.1-pro-preview`
- `gemini-3.5-flash`
- `qwen3.7-plus-preview`
- `qwen3.6-plus`

Open-weight models

- `qwen3.6-27b`
- `qwen3.5-397b-a17b`
- `qwen3-vl-235b-a22b`
- `kimi-k2.6`
- `kimi-k2.5-thinking`
- `gemma-4-31b`
- `gemma-4-26b-a4b`
- `mimo-v2.5`

- `mimo-v2.5-pro`
- `mistral-medium-3.5`
- `mistral-small-4`
- `dots.vlm1`
- `dots.ocr`
- `glm-4.5v-thinking`
- `glm-4.1v-thinking`
- `skywork-r1v3-38b`
- `skywork-r1v2-38b`

1.1 The Procedure

1. Freeze the four leaderboard snapshots: Arena Vision proprietary, Arena Vision open-weight, MMMU-Pro proprietary, and MMMU-Pro open-weight.
2. For each leaderboard, identify the five highest-ranked model families by scanning the ranking from top to bottom. Skip models that cannot be evaluated through a generally available API or downloadable checkpoint. Claude Opus and Claude Sonnet are treated as distinct families because they occupy different capability, cost, and deployment tiers.
3. For each selected family, choose two models:
   (a) the highest-ranked model from that family in the leaderboard. If this model is no longer available through its provider, replace it with the closest currently available model from the same family. For example, Gemini 3.0 Pro appears in the leaderboard, but it is no longer available in Google AI Studio; it is therefore replaced by Gemini 3.1 Pro Preview, accessed through the endpoint `gemini-3.1-pro-preview`;
   (b) the strongest currently available distinct model from the same family. If the strongest current model coincides with the leaderboard-selected model, choose the next strongest currently available distinct model from that family.
4. Collapse variants differing only by inference or serving mode, such as `thinking`, `high`, `instant`, or `flash`, keeping the strongest or highest-ranked variant unless the distinction is itself relevant to the experiment.
5. Remove duplicates across the four leaderboards and evaluate each remaining model once.

This procedure gives at least 10 models.

When a provider exposes a model only through a preview endpoint, the preview endpoint is treated as the implementation of the corresponding leaderboard model, provided that it is the closest available version of the same model family. For example, a leaderboard entry written as Gemini 3.1 Pro is evaluated using `gemini-3.1-pro-preview`.

1.2 The Leaderboards

Model name	Prop. rank	Overall rank	Score	Date	Family	Company
GPT-5.4 Thinking w/ tools	1	1	82.1	2026-03-05	GPT-5.4	OpenAI
GPT-5.4 Thinking w/o tools	2	2	81.2	2026-03-05	GPT-5.4	OpenAI
Gemini 3.0 Pro	3	3	81.0	2025-11-18	Gemini 3	Google DeepMind
Gemini 3.1 Pro Thinking (High)	4	4	80.5	2026-02-19	Gemini 3.1	Google DeepMind
Muse Spark Thinking	5	5	80.4	2026-04-08	Muse Spark	Meta
GPT-5.2 Thinking w/o Python	5	5	80.4	2025-12-11	GPT-5.2	OpenAI
GPT-5.2 Thinking w/o tools	7	7	79.5	2025-12-11	GPT-5.2	OpenAI
GPT-5.1 Thinking	8	8	79.0	2025-11-13	GPT-5.1	OpenAI
GPT-5 w/ thinking	9	9	78.4	2025-08-07	GPT-5	OpenAI
Claude Opus 4.6 w/ tools	10	10	77.3	2026-02-05	Claude Opus 4.6	Anthropic
o3	11	12	76.4	2025-04-16	o3	OpenAI
GPT-5.1	12	13	76.0	2025-11-13	GPT-5.1	OpenAI
Claude Sonnet 4.6 w/ tools	13	14	75.6	2026-02-17	Claude Sonnet 4.6	Anthropic
Claude Sonnet 4.6 w/o tools	14	15	74.5	2026-02-17	Claude Sonnet 4.6	Anthropic
Claude Opus 4.6 w/o tools	15	16	73.9	2026-02-05	Claude Opus 4.6	Anthropic
Claude Opus 4.5	15	16	73.9	2025-11-24	Claude Opus 4.5	Anthropic
Claude Sonnet 4.5	17	20	68.9	2025-09-29	Claude Sonnet 4.5	Anthropic
Gemini 2.5 Pro 05-06	18	22	68.0	2025-05-06	Gemini 2.5	Google DeepMind
Seed 1.5-VL Thinking	19	23	67.6	2025-05-13	Seed 1.5-VL	ByteDance

Table 1.1: Highest-ranked proprietary/API models on the MMMU-Pro leaderboard, extended until five owning companies are represented. Ranks are model-only ranks, excluding human baselines. Tied scores share the same rank.

Model name	Open-weight rank	Overall rank	Score	Date	Family	Company
Gemma 4 31B	1	11	76.9	2026-04-02	Gemma 4	Google DeepMind
Gemma 4 26B A4B	2	18	73.8	2026-04-02	Gemma 4	Google DeepMind
dots.vlm1	3	19	70.1	2025-08-05	dots.vlm	RedNote
Qwen3-VL 235B-A22B	4	21	68.1	2025-09-22	Qwen3-VL	Alibaba Cloud
GLM-4.5V w/ Thinking	5	25	65.2	2025-08-11	GLM-4.5V	Z.ai / Zhipu AI
GLM-4.1V w/ Thinking	6	29	57.1	2025-07-01	GLM-4.1V	Z.ai / Zhipu AI
Skywork-R1V3-38B	7	30	55.4	2025-07-05	Skywork-R1V	Skywork AI

Table 1.2: Highest-ranked open-weight models on MMMU-Pro, extended until five distinct owning companies are represented. Ranks are model-only ranks, excluding human baselines. Seed models are excluded because they are treated here as API-accessible rather than open-weight.

Model name	Prop. rank	Overall rank	Score	Date	Family	Company
claude-opus-4-7-thinking	1	1	1308 ± 8	2026-04-16	Claude Opus 4.7	Anthropic
claude-opus-4-6-thinking	2	2	1303 ± 8	2026-02-05	Claude Opus 4.6	Anthropic
claude-opus-4-7	3	3	1300 ± 8	2026-04-16	Claude Opus 4.7	Anthropic
claude-opus-4-6	4	4	1296 ± 7	2026-02-05	Claude Opus 4.6	Anthropic
muse-spark	5	5	1295 ± 10	2026-04-08	Muse Spark	Meta
claude-opus-4-8-thinking	6	6	1294 ± 13	2026-05-28	Claude Opus 4.8	Anthropic
gemini-3-pro	7	7	1289 ± 8	2025-11-18	Gemini 3	Google
gpt-5.5	8	8	1286 ± 9	2026-04-23	GPT-5.5	OpenAI
gpt-5.4-high	9	9	1283 ± 8	2026-03-05	GPT-5.4	OpenAI
gpt-5.5-high	10	10	1282 ± 9	2026-04-23	GPT-5.5	OpenAI
gpt-5.2-chat-latest-20260210	11	11	1279 ± 7	2025-12-11	GPT-5.2	OpenAI
claude-opus-4-8	12	12	1279 ± 12	2026-05-28	Claude Opus 4.8	Anthropic
gemini-3.1-pro-preview	13	13	1278 ± 6	2026-02-19	Gemini 3.1	Google
claude-sonnet-4-6	14	14	1278 ± 7	2026-02-17	Claude Sonnet 4.6	Anthropic
gpt-5.4	15	15	1275 ± 8	2026-03-05	GPT-5.4	OpenAI
gpt-5.5-instant	16	16	1275 ± 9	2026-04-23	GPT-5.5	OpenAI
gemini-3-flash	17	17	1271 ± 6	2025-12-17	Gemini 3	Google
qwen3.7-plus-preview	18	18	1263 ± 11	2026-05-19	Qwen3.7	Alibaba

Table 1.3: Highest-ranked proprietary models on Arena Vision Overall, extended until five distinct companies are represented.

Model name	Open rank	Overall rank	Score	Date	Family	License	Company
kimi-k2.6	1	19	1261 ± 8	2026-04-20	Kimi K2.6	Modified MIT	Moonshot
gemma-4-31b	2	24	1253 ± 7	2026-04-02	Gemma 4	Apache 2.0	Google
qwen3.5-397b-a17b	3	26	1249 ± 7	2026-02-16	Qwen3.5	Apache 2.0	Alibaba
kimi-k2.5-thinking	4	28	1247 ± 7	2026-01-27	Kimi K2.5	Modified MIT	Moonshot
kimi-k2.5-instant	5	34	1238 ± 11	2026-01-27	Kimi K2.5	Modified MIT	Moonshot
mimo-v2.5	6	35	1238 ± 8	2026-04-27	MiMo-V2.5	MIT	Xiaomi
gemma-4-26b-a4b	7	36	1238 ± 8	2026-04-02	Gemma 4	Apache 2.0	Google
qwen3.5-122b-a10b	8	41	1226 ± 7	2026-02-24	Qwen3.5	Apache 2.0	Alibaba
qwen3.5-27b	9	45	1220 ± 7	2026-02-24	Qwen3.5	Apache 2.0	Alibaba
qwen3-vl-235b-a22b-instruct	10	49	1215 ± 7	2025-09-22	Qwen3-VL	Apache 2.0	Alibaba

Table 1.4: Highest-ranked open-weight models visible in the provided Arena Vision Overall excerpt. The excerpt reaches four distinct companies: Moonshot, Google, Alibaba, and Xiaomi.

1.3 Models Selected per Leaderboard

In the lists below, model names refer to the model identifiers or closest currently available implementation used in the experiments. When a leaderboard entry refers to an unavailable or superseded model, the closest currently available model from the same family is used instead. For example, Gemini Pro entries are evaluated through `gemini-3.1-pro-preview`. Similarly, labels such as `thinking`, `high`, or `w/ tools` are treated as inference or serving configurations rather than separate model families.

Models selected from Table 1.1. The five selected families are GPT, Gemini, Claude Opus, o-series, and Claude Sonnet. The models included from this leaderboard are:

- **GPT**: `gpt-5.4`; `gpt-5.5`.
- **Gemini**: `gemini-3.1-pro-preview`; `gemini-3.5-flash`.
- **Claude Opus**: `claude-opus-4-6`; `claude-opus-4-8`.
- **o-series**: `o3`; `o4-mini`.

- **Claude Sonnet**: `claude-sonnet-4-6`; `claude-sonnet-4-5`.

Muse Spark is excluded because its API is not generally available.

Models selected from Table 1.2. The five selected families are Gemma, dots, Qwen, GLM, and Skywork. The models included from this leaderboard are:

- **Gemma**: `gemma-4-31b`; `gemma-4-26b-a4b`.
- **dots**: `dots.vlm1`; `dots.ocr`.
- **Qwen**: `qwen3-vl-235b-a22b`; `qwen3.6-27b`.
- **GLM**: `glm-4.5v-thinking`; `glm-4.1v-thinking`.
- **Skywork**: `skywork-r1v3-38b`; `skywork-r1v2-38b`.

Models selected from Table 1.3. The five selected families are Claude Opus, Gemini, GPT, Qwen, and Claude Sonnet. The models included from this leaderboard are:

- **Claude Opus**: `claude-opus-4-7`; `claude-opus-4-8`.
- **Gemini**: `gemini-3.1-pro-preview`; `gemini-3.5-flash`.
- **GPT**: `gpt-5.5`; `gpt-5.4`.
- **Qwen**: `qwen3.7-plus-preview`; `qwen3.6-plus`.
- **Claude Sonnet**: `claude-sonnet-4-6`; `claude-sonnet-4-5`.

Muse Spark is excluded because its API is not generally available.

Models selected from Table 1.4. The visible excerpt contains four selected families: Kimi, Gemma, Qwen, and MiMo. The fifth family is obtained by extending the leaderboard beyond the visible excerpt. The models included from this leaderboard are:

- **Kimi**: `kimi-k2.6`; `kimi-k2.5-thinking`.
- **Gemma**: `gemma-4-31b`; `gemma-4-26b-a4b`.
- **Qwen**: `qwen3.5-397b-a17b`; `qwen3.6-27b`.
- **MiMo**: `mimo-v2.5`; `mimo-v2.5-pro`.
- **Mistral**: `mistral-medium-3.5`; `mistral-small-4`.

1.4 Execution Plan for Open-Weight Models

The open-weight models will be divided into two execution groups. Models whose highest-precision weights fit within a single NVIDIA B300 GPU (the GPU in RunPod with the highest available VRAM) will be evaluated locally on RunPod. Models whose highest-precision weights do not fit within a single B300 will be evaluated using hosted inference whenever a suitable hosted endpoint is available.

A single B300 GPU provides approximately 288 GB of VRAM. The VRAM estimates in Table 1.5 refer only to the memory required to store the model weights. They do not include inference overhead, KV cache, image-token memory, batching overhead, CUDA graphs, or framework-specific runtime memory. Therefore, models whose weights are close to 288 GB are treated as borderline even if the raw weight estimate is technically below 288 GB.

For a model with N parameters and b bits per parameter, the weight-only VRAM requirement is estimated as

$$\mathrm{VRAM}_{\mathrm{weights}} \approx N \cdot \frac{b}{8}.$$

For mixture-of-experts models, the estimate uses the total number of parameters, not only the number of active parameters, since the full set of expert weights must normally be available during inference.

Table 1.5: Execution plan for open-weight models under a single-B300 R

Model	Native precision / size	Weight-only VRAM	Execution plan
`dots.ocr`	$\sim$ 2.9B, 16-bit	$\sim$ 5.8 GB	RunPod, single B300
`glm-4.1v-thinking`	$\sim$ 9B, 16-bit	$\sim$ 18 GB	RunPod, single B300
`gemma-4-26b-a4b`	$\sim$ 26B, 16-bit	$\sim$ 52 GB	RunPod, single B300
`qwen3.6-27b`	$\sim$ 27B, 16-bit	$\sim$ 54 GB	RunPod, single B300
`gemma-4-31b`	$\sim$ 31B, 16-bit	$\sim$ 62 GB	RunPod, single B300
`skywork-r1v2-38b`	$\sim$ 38B, 16-bit	$\sim$ 76 GB	RunPod, single B300
`skywork-r1v3-38b`	$\sim$ 38B, 16-bit	$\sim$ 76 GB	RunPod, single B300
`mistral-small-4`	$\sim$ 119B, native FP8	$\sim$ 119 GB	RunPod, single B300
`glm-4.5v-thinking`	$\sim$ 106B, 16-bit	$\sim$ 212 GB	RunPod, single B300
`mistral-medium-3.5`	$\sim$ 128B, 16-bit	$\sim$ 256 GB	RunPod, single B300,
`mimo-v2.5`	$\sim$ 310B, native FP8	$\sim$ 310 GB	Hosted inference or mu
`qwen3-vl-235b-a22b`	$\sim$ 235B, 16-bit	$\sim$ 470 GB	Hosted inference
`kimi-k2.5-thinking`	$\sim$ 1T, native mixed INT4	$\sim$ 550–600 GB	Hosted inference
`kimi-k2.6`	$\sim$ 1T, native mixed INT4/NVFP4	$\sim$ 600–714 GB	Hosted inference
`dots.vlm1`	$\sim$ 672B, FP8 deployment	$\sim$ 672 GB	Hosted inference if ava
`qwen3.5-397b-a17b`	$\sim$ 397B, 16-bit	$\sim$ 794 GB	Hosted inference
`mimo-v2.5-pro`	$\sim$ 1.02T, native FP8	$\sim$ 1020 GB	Hosted inference

Under this plan, the benchmark keeps a common evaluation procedure for all models: the same datasets, prompt templates, image preprocessing, answer extraction code, scoring metrics, and logging format are used regardless of the inference backend. The difference lies only in the execution backend. Models that fit within a single B300 are evaluated locally on RunPod, while larger models are evaluated through hosted inference when available.

This distinction is important methodologically. Accuracy scores may still be compared across the full set of models, provided that the backend, precision, quantization format, and model identifier are recorded for each run. However, latency, throughput, and cost should not be treated as directly comparable between the RunPod-local group and the hosted-inference group, since those measurements depend strongly on the serving infrastructure.

For this reason, each benchmark result should include at least the following metadata:

model identifier	exact checkpoint or provider model name,
backend	RunPod, hosted API, or multi-GPU deployment,
precision	BF16, FP8, INT4, provider-declared, or unknown,
hardware	single B300, hosted endpoint, or cluster,
date tested	date on which the result was obtained.